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Optimization of the Poloidal Magnetic System of the MEPHIST-0 Tokamak

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Abstract—The article is devoted to the calculation of equilibrium configurations of the MEPHIST-0 tokamak plasma with a poloidal magnetic system taking into account the part responsible for compensating the magnetic field of the central solenoid using the DINA plasma physics code. The parameters of the poloidal magnetic system necessary for the formation of initial equilibrium after the discharge breakdown are obtained. The results of calibration experiments without taking into account the vacuum chamber using a pulsed current source are presented, on the basis of which the dynamic problem of plasma equilibrium calculation is formulated. The simulation results are compared with the experimental data, and the influence of the chamber on the results is considered.

Keywords: tokamak, plasma, current, equilibrium

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INTRODUCTION

MEPHIST-0 is a small spherical tokamak being built at the Department of Plasma Physics of the National Research Nuclear University MEPhI. The main objectives of the facility include development of technologies to control plasma–wall interactions, study of microwave pre-ionization, and development of liquid metal applications in tokamaks. The key advantages of the installation include the simplicity of its design, while it has parameters, in particular, the value of the toroidal field, which make it possible to further expand the range of research. The main design parameters of the installation [1]:

Large plasma radius, cm	25
Small plasma radius, cm	13
Aspect ratio	1.9
Plasma elongation	2
Triangularity	0.4
Toroidal field on axis, T	1
Plasma current, kA	160
Discharge duration, ms	30

The analysis of stationary equilibrium plasma configurations in the tokamak makes it possible to determine the parameters of the poloidal magnetic system power sources required for their formation. The stationary configurations are understood as configurations at a certain time without taking into account dynamic processes. By solving the dynamic problem, it is possible to consider the evolution of the plasma throughout the discharge with the prospect of creating an active system for controlling the position and shape of the plasma filament. Currently, the problems of creating stationary and dynamic plasma equilibrium models and their calibration using experimental data are being solved.

In this article, equilibrium limiting plasma configurations at the time immediately after the breakdown are considered. In addition, a model of bringing currents by the power system to computational values in the first approximation is formulated. For these purposes, the dynamic equilibrium problem is solved and verified using calibration experiments on a pulsed current source. The current values optimal in terms of the required number of turns in the poloidal system wind-

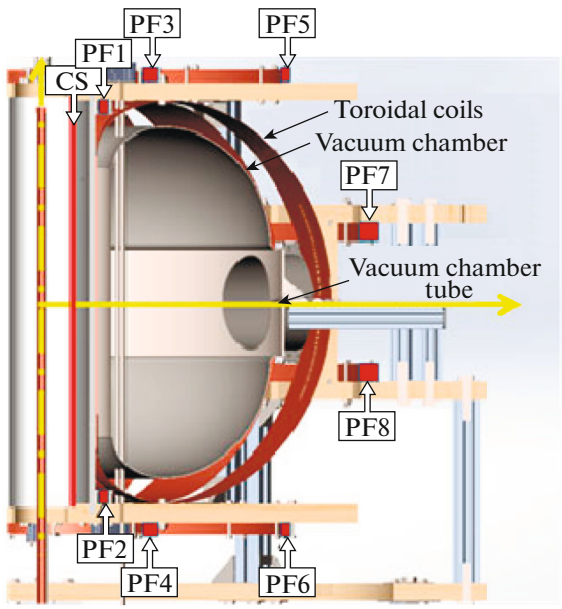

Fig. 1. Schematic representation of the MEPHIST-0 tokamak poloidal system.

ings were selected to minimize the vertical field by interpolation of the experimental data.

The plasma physical DINA code [2] is used to solve the formulated problems.

POLOIDAL SYSTEM OF THE MEPHIST-0 TOKAMAK

The work was carried out taking into account the updated geometrical configuration of the MEPHIST-0 installation, which includes an air central solenoid which is the main source of magnetic flux, poloidal coils for compensating the scattered magnetic fields, and poloidal coils for controlling the position of the plasma filament. The graphic representation of the poloidal system is shown in the schematic section (Fig. 1), where CS is the central solenoid, and PF1–PF8 are the poloidal coils. The coordinates of the centers of the elements of the poloidal system are shown in Table 1.

Each coil of the poloidal magnetic system consists of compensation and control turns. The turns of one coil of both types are located in the same places. The compensation turns are designed to compensate the vertical magnetic field from the inductor in the central area of the vacuum volume. These turns are connected in series with the central solenoid, and accordingly, the currents in them are the same. Together, the central solenoid and compensation turns are the inductor of the installation. This approach to minimizing the vertical field of the inductor was first proposed in [3].

On the other hand, the control turns are directly responsible for controlling the position of the plasma filament in space and have independent power sources. For simplicity, they will be referred to as poloidal turns further in the article.

Only the conductive elements of the vacuum chamber are considered as a passive control system. Since the installation currently has no active control systems and only preprogrammed discharges are planned, the functions of the poloidal system of the tokamak under consideration include the problems of compensating the vertical field of the inductor and providing a given position of the plasma filament.

POLOIDAL SYSTEM MODEL

Using the DINA code, the problem of 2D plasma equilibrium is solved. In this connection, a tokamak is considered as an object model under the assumption of complete axial symmetry of the installation, which includes a conducting vacuum chamber, inductor, and poloidal turns of the magnetic system. The chamber model is represented as a set of 24 toroidal conductors in which current is induced during the plasma discharge. The poloidal field coils are placed in the model according to Table 1; the dimensions of each of the coils depend on the number and position of turns in them. A turn is made of a wire of rectangular cross section (2 × 8 mm). All coils, except for the central solenoid, have two independent parts, one of which refers to the inductor (eight coils and a solenoid) and the other to the main poloidal part (eight coils) of the poloidal magnetic system. Therefore, the magnetic system model includes 17 separate poloidal field coils.

During the calibration experiments designed to test and tune the software model, current was applied to the coils from a pulsed source (a capacitor with a capacity of 5600 μF) with a voltage of up to 3 kV. Different configurations of the poloidal system are considered in the work (in particular, the number of turns in each coil varies). If an experiment was performed for the configuration of the poloidal system under consideration, the experimental current dependence $I(t)$ obtained with the Rogowski belt mounted at the inductor output was used. Otherwise, in the absence of experimental data, the current dependence (Fig. 2)

Table 1. Position of coil groups of the poloidal system of the MEPHIST-0 tokamak

Coordinate	PF1	PF2	PF3	PF4	PF5	PF6	PF7	PF8	CS
r, cm	10	10	17	17	40	40	53	53	5.1
z, cm	34	−34	37	−37	37	−37	18.5	−18.5	±30

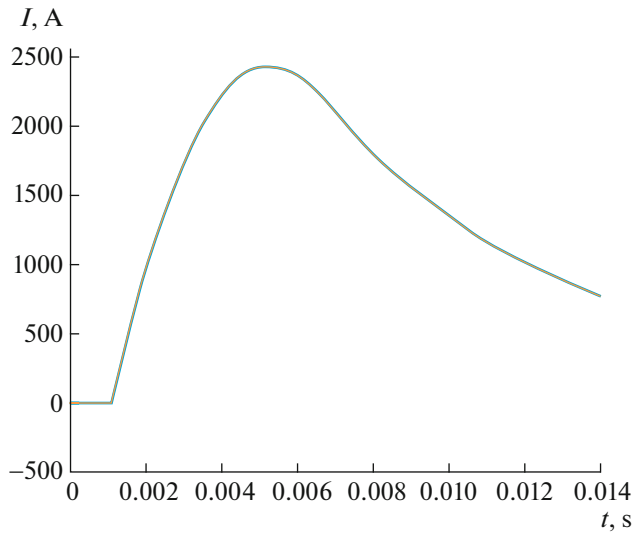

Fig. 2. Reference graph of the current applied to the poloidal coils: (——) experiment and (——) calculation.

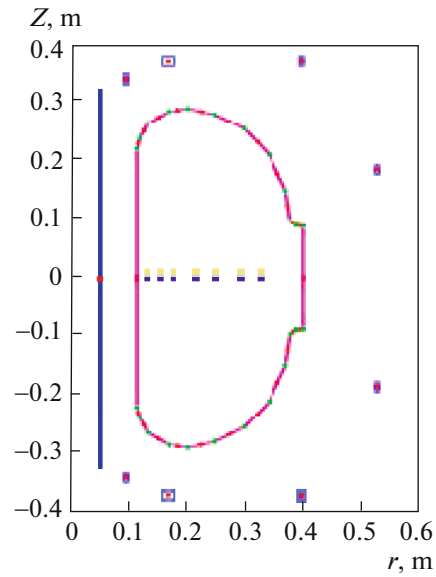

Fig. 3. Model of the poloidal system used in the DINA code.

taken from the Rogowski belt and obtained in the experiment when charging the capacitor to 2 kV using the configuration of the poloidal system including only the inductor coils was used for the considered configuration.

During the experiments, the magnetic field in the installation is measured by seven Hall sensors mounted along the horizontal axis in the meridional cross section of the tokamak and configured to measure the vertical component of the magnetic field. In order to compare the simulation and experimental results, seven sensors were included in the model arranged according to the data in Table 2, which correspond to the positions of the sensors in the installation. All elements of the poloidal magnetic system model as well as the Hall sensors are shown in Fig. 3.

BRIEF MATHEMATICAL DESCRIPTION OF THE CODES USED IN THIS WORK

Equilibrium Configurations

The DINA-SVD module [4] was used to numerically calculate the plasma configuration. The underlying model is based on the solution of the Grad–Shafranov equation in the axisymmetric plasma representation with a free boundary with the expression of toroidal plasma current density component (2):

$$\frac{\partial^2\Psi}{\partial z^2}+\frac{\partial^2\Psi}{\partial r^2}-\frac{1}{r}\frac{\partial^2\Psi}{\partial z^2} = \begin{cases} -2\pi\mu_0 r j_\varphi \to (r,z)\in S, \\ -2\pi\mu_0\sum\limits_{i=1}^{L} r_i I_i \delta(r-r_i)\delta(z-z_i) \to (r,z)\notin S; \end{cases} \quad (1)$$

$$j_\varphi = \left(2\pi r\frac{dp}{d\Psi}+\frac{\mu_0}{4\pi r}\frac{dF^2}{d\Psi}\right). \quad (2)$$

Equations (1) and (2) relate the poloidal magnetic flux $\Psi(r, z)$ to the distribution of plasma pressure $p(\Psi)$ and poloidal current $F(\Psi)$. Here, L is the number of current-carrying conductors; S is the area of the poloidal section occupied by plasma; and r and z are the radial and vertical spatial coordinates. In the case of stationary equilibrium, only conductors of the active system are taken into account ($L = 17$); in the case of the dynamic problem solution, more conductors modeling the chamber $N = 24$ for the dynamic problem $L = 41$ are added.

The algorithm for solving this problem consists in solving Eq. (1) and determining the values of currents I_i ($i = 1, ..., L$) in conductors. In this case, the system of equations is overdetermined, i.e., such that the number of equations in the system exceeds the number of unknowns. This problem is called the *best approximation problem*, has a single solution in the class of

Table 2. Hall sensor positions on the horizontal axis of the installation

Coordinate	Sensor 1	Sensor 2	Sensor 3	Sensor 4	Sensor 5	Sensor 6	Sensor 7
r, cm	13.5	15.7	18.0	22.0	25.3	29.5	32.8

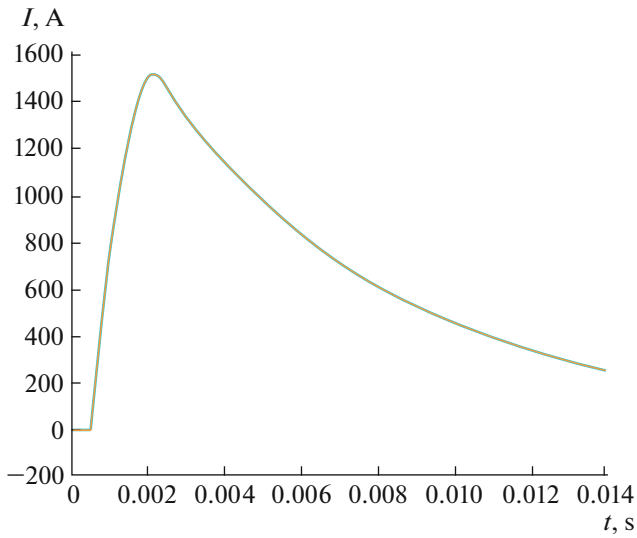

Fig. 4. Input current from a pulsed current source: (——) experiment and (——) calculation.

applied constraints, and is solved numerically by the model of Singular Value Decomposition (SVD) [5].

Dynamic Problem

In this work, we considered the dynamic problem of calculating the magnetic field configuration in the absence of plasma, the results of the solution of which were compared with the experimentally obtained data for the subsequent fine-tuning of the tokamak model. The solution of this problem was used in the calibration experiments to determine the requirements for the power sources. In order to solve it, we used the SIMULINK model created on the basis of the DINA code, which is a simplified version of a similar code used to solve the problem of plasma equilibrium evolution in the T-15MD tokamak [6]. The approach is based on a linearization of the poloidal system model. The linear model is defined by a set of states, in this case current increments,

$$\delta\mathbf{x} = \mathbf{I}(t) - \mathbf{I}_0, \tag{3}$$

defining the vector $\mathbf{x}$, the dimension of which is equal to the number of elements of the poloidal system (17 active coils and 24 elements of the conducting chamber); $\mathbf{I}_0$ is the vector of initial currents.

The currents are related to the poloidal magnetic flux by Kirchhoff's law

$$\frac{d\mathbf{\Psi}}{dt} + R\mathbf{I} = \mathbf{U}, \tag{4}$$

where R is the resistance matrix. Further, by linearizing the system of equations (4), we create a linear model of the object

$$\begin{cases} \dfrac{d}{dt}(\delta\mathbf{x}) = A\delta\mathbf{x} + B\delta\mathbf{U}; \\ \mathbf{y} = C\delta\mathbf{x} + D\delta\mathbf{U}, \end{cases} \tag{5}$$

where the vector $\mathbf{y}$ characterizes the plasma shape parameters. A detailed process for obtaining this system was described in [7]. Linear system matrices A, B, C, D are constructed using known parameters of the device, including geometric parameters and resistances of chamber elements and poloidal system coils, and positions of the limiter, magnetic probes, and loops.

CALIBRATION EXPERIMENTS

Calibration experiments are necessary to verify the used tokamak model and sensors. The experiments were performed with each poloidal coil connected in series. During the experiments, a current was supplied to the coils of the poloidal system with a pulsed current source, which was a capacitor with a capacitance of 5600 μF charged to 400 V. Further, the dependences of the values of the vertical magnetic field obtained from the Hall sensors installed on the installation axis were compared with the calculated values. Since the experiments were carried out with the vacuum chamber removed from the installation, the dependences were obtained without taking into account the influence of currents from the vacuum chamber elements.

Figure 4 shows the current evolution during the discharge when only the PF5 coil is connected. Figure 5 compares the calculated vertical magnetic field with the experimental dependence obtained on the fourth Hall sensor. Here, blue is the experimental data, black is the calculation without the chamber, and red is the calculation with the chamber.

From the data in Fig. 5 we can conclude that it is possible to use the created installation model in further calculations, the time positions of the peaks on the experimental and calculated (without taking the chamber into account) plots almost coincide. In some experiments, we observe deviations of the amplitude value to the lower side within $\delta = 15\%$ with preservation of the equality of the time positions of the peaks and the form of the time dependence. Deviations can be explained by the bias, which includes the inaccuracy of magnetic field probe location, losses in the electric circuit, and inaccurate impedance data for each of the coils.

If we compare the calculated curve taking into account the vacuum chamber (red) and without taking it into account (black), we can note the time shift of the peak associated with the effects of mutual induction, while the type of curve does not change.

Figures 6 and 7 contain, respectively, diagrams of the magnetic field distribution and the difference between the experimental and calculated (without taking into account the chamber) results using data

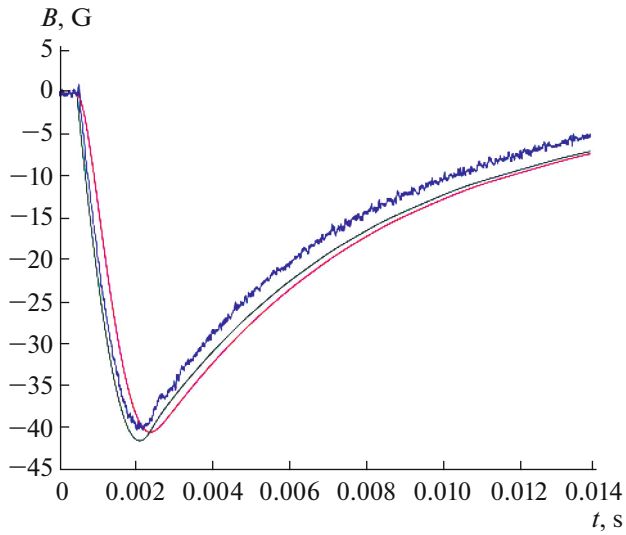

Fig. 5. Comparison of the calculated field with experimental data for Hall sensor 7: (——) PF coils + chamber; (——) PF coils; and (——) experiment.

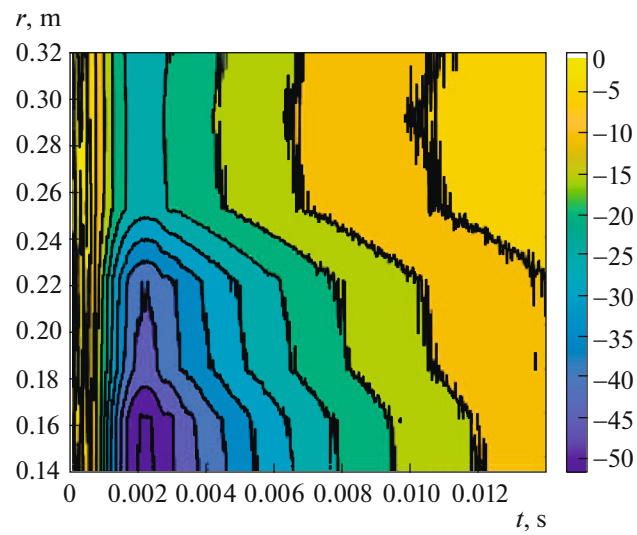

Fig. 6. Experimentally obtained vertical magnetic field distribution.

from seven Hall sensors with linear interpolation of the field magnitude signals between the sensors for the calibration experiment under consideration. Here, the color indicates the magnitude of the field, and the vertical axis corresponds to the position on the horizontal axis of the installation.

The diagram in Fig. 7 shows that the calculation error is approximately the same for all seven probes. Therefore, it is possible to talk about the correct modeling of the vertical magnetic field in the chamber. On average, the difference for the discharge in question at the peak (corresponding to 0.002 s in Fig. 5) is 18% between the calculated value of the magnetic field without taking into account the vacuum chamber and the experimental data.

SIMULATION OF THE OPTIMAL CONFIGURATION

Inductor

This article considers a variant of the inductor consisting of a series-connected central solenoid and compensation turns of the tokamak's poloidal magnetic system. The main goal of the optimal configuration of the poloidal system is to create a minimum vertical field in the area of the large radius of the tokamak $R = 25$ cm, which is a necessary condition for the breakdown.

This configuration was chosen by interpolating the data in the measured configurations (more than 40). The experiments were performed according to experiment planning methods [8]; the input factor was the number of turns in each of the coils under consideration, and the response function was the magnetic field. At the first stage of the study, the exhaustive search of the combinations of the number of turns and current directions corresponded to the plan of a full-factorial two-level experiment, in which the number of experiments was determined by the condition $N = 2^4$. Since a two-level experiment was chosen, each factor can only take a maximum and a minimum value, i.e., an upper and a lower level, respectively. The base of degree 2 meets this condition. Degree 4 represents the number of factors according to the number of transformer coils with a variable number of turns. During the experiments, the upper and lower values are set in all possible combinations. Therefore, the number of

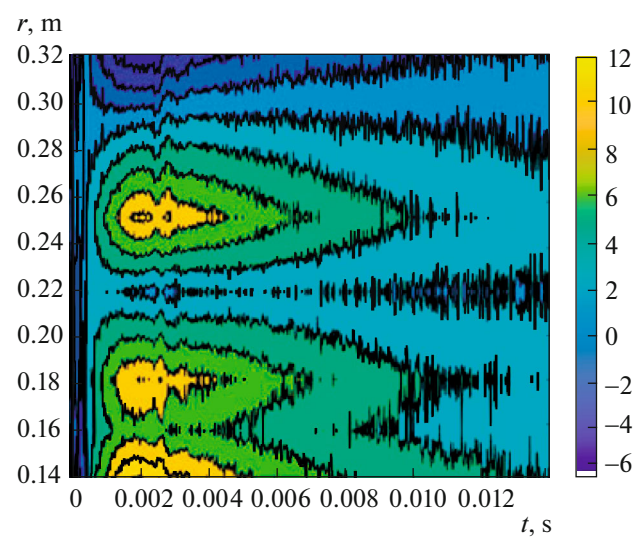

Fig. 7. Difference between the calculated and experimental values of the field.

Table 3. Optimal inductor configuration

Coil	Number of turns
PF1/2	−1
PF3/4	1
PF5/6	1
PF7/8	0
CS	75

experiments to obtain an incomplete quadratic model is $N = 16$.

However, in the investigated area of the factor space, a pronounced nonlinearity was found; therefore, in order to increase the degree of the resulting polynomial, experiments were additionally conducted according to the plan of a fractional-factor three-level experiment according to the plan 3^{4-1}. Here, the values of the factors were varied at three levels: maximum, average, and minimum. Since the number of experiments would be $N = 81$ if the matrix of the full-factorial three-level experiment were used, a system of experiments representing its part was taken, a fractional replica, where the degree of the number 4 − 1 is an indicator of the fractionality of the plan.

Therefore, using regression and statistical analysis, a mathematical model was obtained from the experimental data, which is a segment of the Taylor series, in which the function of the dependence of the magnetic field on the number of turns is expanded:

$$y = b_0 + \sum b_i x_i + \sum b_{ij} x_i x_j + \sum b_{ii} x_i^2 + \sum b_{iij} x_i^2 x_j + \sum b_{ijj} x_i x_j^2 + \ldots, \quad (6)$$

where b_0 is the free term of the regression equation, b_i are linear effects, b_{ii} are quadratic effects, b_{ij} are effects characterizing interactions of input factors among themselves, and $x_{i,j}$ are values of input factors.

The adequacy of the model was tested using Fisher's criterion. In addition, in order to improve the accuracy and test the predictive ability of the obtained model, studies were conducted at a number of control points. Using the obtained approximation formula, the optimal configuration presented in Table 3 was predicted. Here, the names of the coils and the number of turns are given. The number of turns is the same because there is symmetry of the installation relative to the horizontal axis and the ith and jth coils, respectively, have the same radius and are located at the same height above and below the axis. The minus sign in front of the number of turns means that the current direction in the coil is reversed.

As an input parameter, the dependence $I(t)$ obtained for a configuration with a similar load (total number of used turns) during a pulse with the current evolution shown in Fig. 2 was taken.

Figure 8 shows the calculated values of the magnetic fields on the magnetic sensors. Figure 9 shows the distribution of the magnetic field along the radius. It can be seen that, in the region close to the large radius $r = 25$ cm, there is a field zero according to the calculations. Similar results were obtained for the T-9 tokamak [3].

The calculation was carried out with a peak current value of 2.5 kA. If the current amplitude is changed, the position of the field minimum will not shift, because when the currents are varied proportionally, only the amplitude of each of the curves changes.

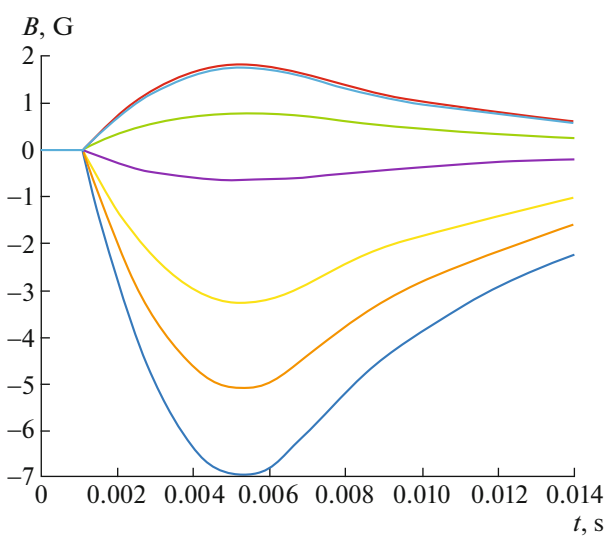

Fig. 8. Calculated vertical magnetic field distribution on Hall sensors 1–7: (——) sensor 1; (——) sensor 2; (——) sensor 3; (——) sensor 4; (——) sensor 5; (——) sensor 6; and (——) sensor 7.

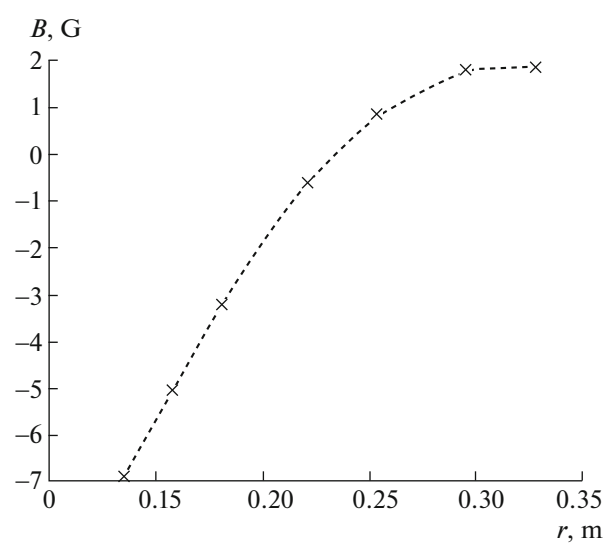

Fig. 9. Distribution of the vertical component of the magnetic field along the radius of the installation.

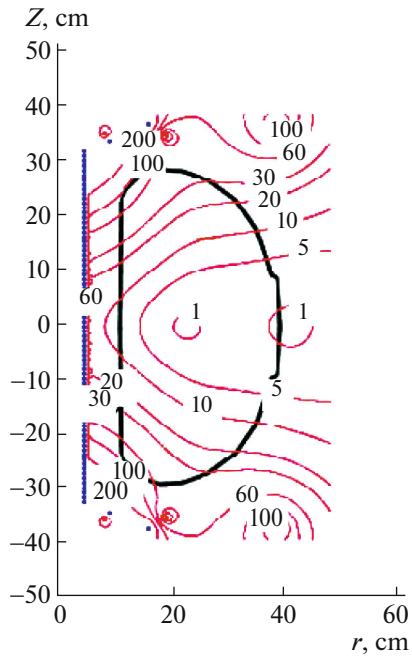

Fig. 10. Distribution of the vertical component of the magnetic field from the inductor when simulated by the optimal configuration shown in Table 4.

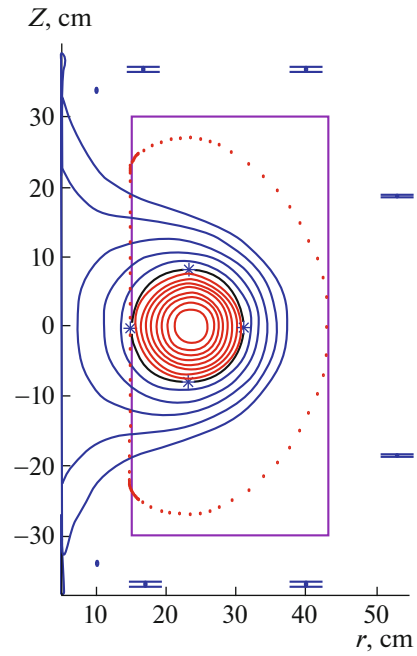

Fig. 11. Distribution of magnetic surfaces when using the equilibrium configuration specified in Table 5.

This fact will be used further in the calculation of stationary equilibrium plasma configurations.

On the basis of this configuration, a 2D model of the distribution of magnetic fields from the inductor of the poloidal system was constructed. The result of the calculations is shown in Fig. 10. The model was built for the maximum current achieved during the pulse.

Full Poloidal System

In addition to the already calculated inductor, let us add poloidal coils responsible for controlling the position of the plasma filament. We calculate the currents in the coils required to form the plasma equilibrium at a baseline plasma current of 160 kA. We consider the limiting configuration on the inner wall of the chamber formed immediately after the breakdown [9]. The effect of the chamber is not taken into account, because stationary equilibrium is considered. Figure 11 shows the distribution of magnetic field lines for the corresponding equilibrium. Here, the red dots mark the geometric boundaries of the chamber, the purple rectangle is the calculated region, the red lines are the magnetic field lines inside the plasma filament, and the blue lines are those outside. The black line indicates the plasma boundary.

The number of turns in the coils is obtained by interpolating the data of 27 calculated equilibria carried out with a linear regression in much the same way as the considered optimal configuration of the transformer system. The purpose of this optimization is to minimize the currents required to maintain equilibrium in the steady-state mode. The final configuration of the optimal poloidal system is given in Table 4. As before, the minus sign before the number of turns means the reverse direction of current in these coils.

During the search for the optimal configuration, it was found that the influence of the number of turns in the coils PF1/2 on the values of currents is insignificant, and in order to simplify the electromagnetic system, they can be dropped. The best results are obtained with an equal number of turns in the coils PF5/6 and PF7/8, and the more there are of them, the less the required currents. Therefore, the number of turns in these coils is taken as the maximum possible for the installation.

The final values of currents obtained for equilibrium in the optimal full configuration of the tokamak poloidal system are given in Table 5.

Table 4. Full configuration of the poloidal system

Coil	Number of turns	
	Inductor	Poloidal turns
PF1/2	–1	0
PF3/4	1	–20
PF5/6	1	–20
PF7/8	0	–5
CS	75	0

Table 5. Rated currents in the inductor and poloidal system in the configuration shown in Table 4

Coil	Rated current I, kA
Inductor	0.22
PF1/2	0
PF3/4	3.91
PF5/6	6.26
PF7/8	0.53

Supply System Calculation

Let us check the existing supply system and, if necessary, calculate the voltages required to form a plasma configuration at the time of breakdown. For this purpose, a modified version of the dynamic equilibrium calculation program described earlier is used, but with allowance for the power system, in particular, the capacitor parameters and the impedance in each coil. The impedance was measured with an RLC meter.

We use the equilibrium obtained earlier for the configuration shown in Table 5. Let us assume that in 0.004 s the power system should provide the currents in the coils necessary to form the plasma configuration at the time of breakdown. This time is taken equal to the time of current rise as the capacitor discharges. We can also assume this time as the time to discharge, because it is approximately 15% of the full planned discharge time. We also assume that the current rise follows a parabolic law, for which an appropriate voltage is selected.

It was determined that the highest voltage to fulfill the programmed current rise is required on the coils PF3/4. For them, the graphs of dependence $I(t)$ for different capacitor voltages are given accordingly. Figure 12 shows that, in order to comply with the calculated model, the voltage on the coil PF3/4 should be more than 2.5 kV, which is feasible with the use of existing capacitors (3 kV limit). Such conclusion confirms the full coincidence of the black (expected dependence) and the green (corresponding to the current dependence at 2.5 kV) lines. Moreover, even at 2 kV (red line), the dependence $I(t)$ coincides with the expected dependence, except for a small section at a time greater than $t = 3.7 \times 10^{-3}$ s.

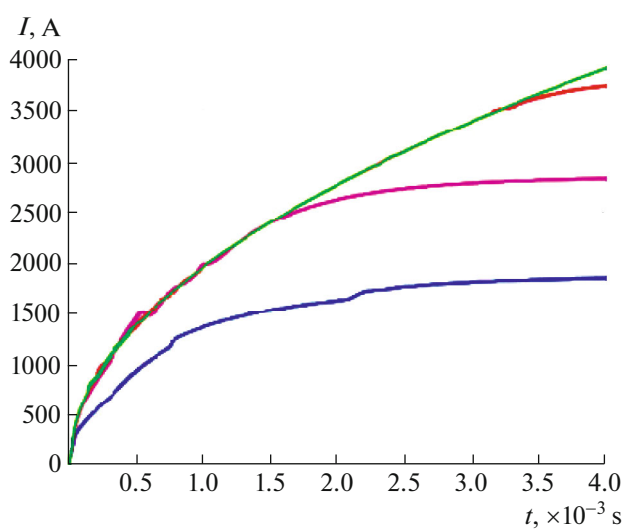

Fig. 12. Currents in coils PF3/4 at different capacitor voltages: (——) expected; (——) 1.0; (——) 1.5; (——) 2.0; (——) 2.5 kV.

Coincidence with the calculated model on the remaining coils is achieved at lower voltages. Therefore, it can be assumed that the power system of the poloidal system of the MEPHIST-0 tokamak has sufficient reserve to provide the necessary currents to create equilibrium at plasma current I_{pl} = 160 kA. It should be noted that this is only a preliminary estimate; accurate calculations require an evolutionary model of the plasma in the discharge.

CONCLUSIONS

In this work, the poloidal system of the MEPHIST-0 tokamak was simulated. The system includes the inductor and the main poloidal system responsible for controlling the position of the plasma filament as well as the conducting vacuum chamber. Using the results of calibration experiments on the poloidal system without the chamber, the model showed good agreement with the results obtained in the experiments on the real installation.

The configuration of the inductor of the poloidal system providing a minimum of the vertical component of the magnetic field in the plasma breakdown region was determined by the interpolation of the experimental data.

The limiting plasma configuration was simulated taking into account the formation of the plasma filament on the interior of the vacuum chamber at the base value of the current I_{pl} = 160 kA and the value of the toroidal magnetic field B = 1 T. The applicability of the existing MEPHIST-0 tokamak supply system for creating such a discharge was also tested.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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